

Question 2 — English (5 marks)

Answered out of order — Q2 first, as permitted.

I disagree with the popular view. I think easy access to information has, on balance, made students worse learners, and I want to argue that carefully rather than just assert it.

Learning happens when you hold a difficulty in your head long enough to resolve it. That process is uncomfortable, and every shortcut out of the discomfort is taken. Before search engines, a student stuck on a problem had no option but to reread, reason and ask. Now the exit is one tap away, and the exit is always taken first. What is practised is retrieval of somebody else's reasoning, not the construction of your own.

I have watched this in my own class. When we were given a set of unseen problems, most of us including me found worked solutions online within minutes and copied the method. In the test two weeks later, where nothing could be looked up, the same problems were much harder than they should have been. We had read the solutions; we had not learned to produce them.

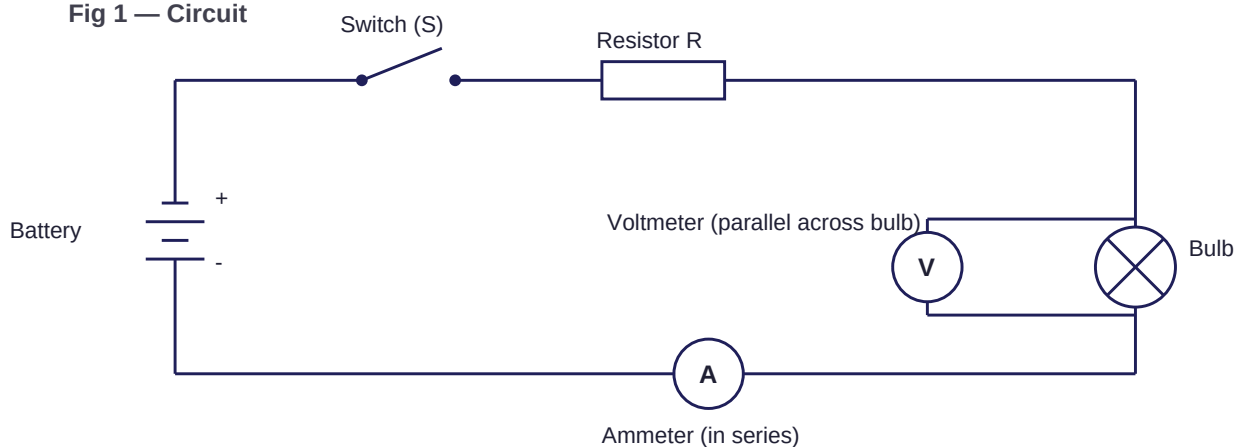
The strongest argument against my position is that technology gives access to students who would otherwise have none — a student without a good teacher can now find an excellent explanation for free. I accept that this is a genuine and important gain, and for a motivated student it is transformative. But that student is exactly the one who would have found a way regardless. For the ordinary student, the same tool mostly removes the need to think.

So my conclusion is that the technology is not neutral in practice, even if it is neutral in principle. It reliably reduces the amount of struggle in a student's day, and since struggle is where learning comes from, it has made most students worse learners even while making them better informed.

Question 1 — Science (5 marks)

A circuit is a closed loop of conductors. The battery supplies the potential difference, the switch completes or breaks the loop, and current flows through the resistor and the bulb only when the loop is complete. The ammeter is in the loop itself, in series, because the same current flows through every series component. The voltmeter goes across the bulb in parallel, since it measures the difference in potential between the bulb's two ends.

Increasing the resistance reduces the current, because for a fixed voltage $I = V/R$.

Fig 1 — Circuit**Question 3 — Economics (5 marks)**

Price is on the vertical axis, quantity on the horizontal. Demand slopes downward, supply slopes upward, and the two cross at Rs 30 and 60 units. At that price quantity demanded equals quantity supplied, which is why it is the equilibrium.

Below Rs 30 buyers want more than sellers will supply, so there is a shortage and the price is bid up. Above Rs 30 sellers bring more than buyers want, so there is a surplus and the price is cut. In both cases the market is pushed back towards Rs 30.

A rise in production costs makes supplying less profitable at every price, so the supply curve shifts left. The new equilibrium sits higher up the demand curve: a higher price and a smaller quantity traded.

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Fig 2 — Market equilibrium and a leftward supply shift

